



## 金属结构设计计算书 Calculation Sheet of Structure Design

### 1.起重机的设计要求

### 1. Crane Design Requirements

#### 1.1.基本参数

#### 1.1 Basic Parameters

起重量:  $Q=18t$  跨度  $L=37.1m$  起升高度  $H=30m$

Rated hoisting load:  $Q=18t$  Span:  $L=37.1m$  Hoisting height:  $H=30m$

起升速度  $v_q=50m/min$ , 起、制动时间  $3s$ , 加速度  $a_q=0.28m/s^2$

Hoisting speed:  $v_q=50m/min$ , starting and braking time:  $3s$ , acceleration:  $a_q=0.28m/s^2$

大车运行速度  $v_y=70m/min$ , 起、制动时间  $6s$ , 加速度  $a_y=0.20m/s^2$

Crane speed:  $v_y=70m/min$ , starting and braking time:  $6s$ , acceleration:  $a_y=0.20m/s^2$

小车运行速度  $v_{yt}=55m/min$ , 起、制动时间  $4s$ , 加速度  $a_y=0.23m/s^2$

Trolley speed:  $v_{yt}=55m/min$ , starting and braking time:  $4s$ , acceleration:  $a_y=0.23m/s^2$

小车运行到左极限位置时, 抓斗中心线到大车轨道中心线的距离为  $L_{\min}=1.850m$  (左极限), 大车基距  $B=2.0m$ , 小车基距  $b=2.4m$ 。

When the trolley reaches the left limit position, the distance from the grab centerline to the centerline of the crane rail is  $L_{\min}=1.850m$  (left limit), the base of the crane is  $B=2.0m$ , and the base of the trolley is  $b=2.4m$ .

整机工作级别 A8

Working class of crane A8

起升机构级别: M8

Class of hoisting mechanism: M8

起升机构的起升状态级别为 HC2, 起升驱动装置可作稳定低速运转。

The hoisting status level of the hoisting mechanism is HC2, and the hoisting driving device can be operated stably at low speed.

大车运行机构级别: M8

Class of crane travelling mechanism: M8

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小车运行机构级别: M8  
Class of trolley travelling mechanism: M8

金属结构级别: M8  
Class of metal structure: M8

## 1.2.结构计算遵循的规范

## 1.2 Specifications Followed by Structural Calculation

EN 13001-1: 2015 Cranes-General design - General principles and requirements

EN 13001-2: 2021 Crane safety - General Design-Load actions

DIN15080-1: Cranes Steel structures Verification and analyses

## 1.3.结构设计的许用值

## 1.3 Allowable Value of Structural Design

### 1.3.1.材料的许用应力

### 1.3.1 Allowable Stress of Materials

起重机结构件的许用应力按式 (1-1) 计算, 并列于表 1.1。

The allowable stresses of crane structural parts are calculated according to Formula (1-1) and are listed in Table 1.1.

$$\sigma_i \leq [\sigma]_i = \frac{R}{\gamma_f \gamma_n} \quad (1-1)$$

式中:  $R$ ——强度的标准值。对于屈强比 $<0.7$ 的钢材,  $R = \sigma_s$ ; 对于屈强比 $\geq 0.7$ 的高强度钢材,  $R = 0.5\sigma_s + 0.35\sigma_b$ 。

In the formula:  $R$  - standard value of strength. For steels with yield strength to tensile strength ratio  $<0.7$ ,  $R = \sigma_s$ ; for high-strength steels with the yield strength to tensile strength ratio of  $\geq 0.7$ .

$\sigma_s$ ,  $\sigma_b$  ——钢材屈服强度和抗拉强度, MPa;

$\sigma_s$ ,  $\sigma_b$  - Yield strength and tensile strength of steel, MPa;

$\gamma_f$  ——强度系数, 按表 1.1 选择;

$\gamma_f$  - Strength coefficient, selected according to Table 1.1;

$\gamma_n$  ——高危系数, 取  $\gamma_n = 1.1$ 。

$\gamma_n$  - High-risk coefficient,  $\gamma_n = 1.1$ .

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表 1.1 强度系数  
Table 1.1 Strength Coefficient

|                              |      |      |      |
|------------------------------|------|------|------|
| 工况<br>Operating condition    | A    | B    | C    |
| 强度系数<br>Strength coefficient | 1.48 | 1.34 | 1.22 |

表 1.2 各工况下的许用应力  
Table 1.2 Allowable Stress under Each Working Condition

| 材料名称<br>Material Type | 板厚<br>Plate Thickness<br>mm | 屈服强度<br>Yield Strength<br>MPa | 抗拉强度<br>Tensile Strength<br>MPa | 屈服比<br>Yield to tensile Ratio | 许用应力 $[\sigma]$ , MPa<br>Allowable Stress $[\sigma]$ , MPa |                             |                             |
|-----------------------|-----------------------------|-------------------------------|---------------------------------|-------------------------------|------------------------------------------------------------|-----------------------------|-----------------------------|
|                       |                             |                               |                                 |                               | 工况 A<br>Working Condition A                                | 工况 B<br>Working Condition B | 工况 C<br>Working Condition C |
| S355JR                | ≤16                         | 355                           | 470~630                         | ≤0.7                          | 218                                                        | 240                         | 264                         |
|                       | >16~40                      | 345                           |                                 |                               | 211                                                        | 234                         | 257                         |
|                       | >40~63                      | 335                           |                                 |                               | 205                                                        | 227                         | 249                         |

### 1.3.2. 设计的许用刚度 1.3.2 Allowable Stiffness of the Design

结构的静刚度以结构在规定载荷作用下指定截面的静态挠度倒数 $1/Y_L$ 来表征。静刚度的设计准则为：

The static rigidity of the structure is characterized by the reciprocal  $1/Y_L$  of the static deflection of the specified section under the specified load. The design criteria of static rigidity is:

$$Y_L \leq [Y_L] = 37100/1000 = 37.1\text{mm}$$

式中： $Y_L$ ——垂直静挠度； $[Y_L]$ ——许用挠度值，跨中按 $S/1000$ 计，其中 $S$ 为跨度。

In the formula:  $Y_L$  - vertical static deflection;  $[Y_L]$  - allowable deflection value, the mid-span shall be calculated according to  $S/1000$ , wherein  $S$  is the span.

对于桥式类型起重机，当满载小车位于跨中有效工作位置时，主梁的对应截面在移动载荷（额定起升载荷和小车自重载荷之和，不计动载系数）作用下的挠度值。  
For the bridge crane, when the trolley with full load is located at the effective working position in the span, the corresponding section of the main beam is deflected under the action of moving load (the sum of the rated hoisting load and the self-weight load of the trolley, excluding the dynamic load coefficient).

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## 2. 载荷组合和计算载荷 2. Load Combination and Calculation Load

### 2.1. 载荷组合 2.1 Load Combination

#### 2.1.1. 载荷工况描述 2.1.1 Description of Load Conditions

表 2.1 结构计算载荷工况  
Table 2.1 Structural Calculation Load Conditions

| 载荷工况<br>Load Conditions | 起重机作业状态描述<br>Description of Crane Operation State                                                                                                                                                |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| A1                      | 大车不动，小车运行至规定位置（跨中或端部极限位置）起升或下降载荷<br>The crane is not operated, and the trolley travelled to the specified position (mid-span or end limit position) for hoisting or lowering load.               |
| A2                      | 大车运行制动，小车位于规定位置起升或下降载荷<br>The crane is being braked, and the trolley is located at the specified position for hoisting or lowering load.                                                         |
| A3                      | 小车位于跨中满载起升或下降制动，同时大车平稳制动<br>The trolley is in the span with full load hoisting or lowering braking, while the crane is in stable braking.                                                        |
| B1                      | 小车位于跨端极限位置满载下降制动，同时大车稳定运行<br>The trolley is at the limit position of the span end, and the crane is in stable operation.                                                                         |
| B2                      | 小车位于跨端极限位置满载下降制动，同时大车偏斜运行<br>The trolley is at the limit position of the span end, and the crane is in eccentric operation.                                                                      |
| C1                      | 小车位于跨中，平稳起吊静态试验载荷，小车和大车平稳制动<br>The trolley is located in the middle of span, sable hoisting and static test load, and the trolley and crane are in stable braking.                               |
| C2                      | 小车位于跨中，起吊动态试验载荷起升或下降制动，同时大车运行制动<br>The trolley is located in the middle of span with hoisting or lowering braking at the dynamic test load during the hoisting, while the crane is being braked. |

#### 2.1.2. 计算载荷组合 2.1.2 Calculation Load Combination

桥式起重机的桥架结构主要由主梁和端梁构成。计算的载荷组合见表 2.2。  
The bridge structure of bridge crane is mainly composed of main beam and end beam.  
The calculation load combination is as shown in Table 2.2.

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表 2.2 计算载荷组合  
Table 2.2 Calculation Load Combination

|                                                           | A 类<br>Class A |          |          | B 类<br>Class B |          | C 类<br>Class C |              |  |
|-----------------------------------------------------------|----------------|----------|----------|----------------|----------|----------------|--------------|--|
|                                                           | A1             | A2       | A3       | B1             | B2       | C1             | C2           |  |
| 桥架自重载荷 DL<br>Self-weight load DL of bridge frame          | $\phi_1$       | $\phi_1$ | $\phi_4$ | $\phi_1$       | $\phi_4$ | $\phi_1$       | $\phi_4$     |  |
| 移动载荷 TL<br>Moving load TL                                 | $\phi_1$       | $\phi_1$ | $\phi_4$ | $\phi_1$       | $\phi_4$ | $\phi_1$       | $\phi_4$     |  |
| 起升载荷 LL+LS<br>Hoisting load LL+LS                         | $\phi_2$       | $\phi_2$ | $\phi_2$ | $\phi_2$       | $\phi_2$ | 1.25           | 1.1 $\phi_6$ |  |
| 小车运行惯性载荷 LATT<br>Inertial load of trolley travelling LATT | $\phi_5$       |          |          | $\phi_5$       |          | 1              |              |  |
| 大车运行惯性载荷 LATG<br>Inertial load of crane travelling LATG   |                | 1        | $\phi_5$ | $\phi_5$       | $\phi_5$ | 1              | $\phi_5$     |  |
| 大车运行偏斜载荷 SKW<br>Eccentric load of crane travelling SKW    |                |          |          |                | 1        |                | 1            |  |

## 2.2.计算载荷 2.2 Calculation Load

### 2.2.1 动载系数 2.2.1 Dynamic Load Coefficient

(1) 起升冲击系数  $\phi_1$  :

(1) Hoisting impact coefficient  $\phi_1$  :

当物品离地起升时，或将悬吊在空中的部分物品突然卸除时，或悬吊在空中的物品下降制动时，起重机本身（主要是对金属结构）的自重因出现振动而产生脉冲式增大或减小的动力响应。取  $\phi_1=1.1$ 。

When the load is hoisted from the ground, or when the hoisted load is suddenly removed, or when the hoisted load is lowering braking, the weight of the crane itself (mainly on the metal structure) will produce the dynamic response of pulse increase or decrease due to the vibration.  $\phi_1=1.1$ .

(2) 起升冲击载荷系数  $\phi_2$  :

(2) Hoisting impact load coefficient  $\phi_2$  :

当物品无约束地起升离地时，物品的惯性力会使起升载荷出现动载增长的作用。按起升状态级别 HC2，则有

When the hoisting object is hoisted off the ground without restraint, the inertia force of the object will cause the hoisting load to increase in the dynamic load. If the

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hoisting state level is HC2,

$$\phi_2 = \phi_{2\min} + \beta_2 v_q = 1.1 + 0.34 \times 50/60 = 1.38$$

式中:  $\phi_{2\min}$  ——与起升状态级别相对应的起升动载系数的最小值, 取为 1.1。

In the formula: the minimum value of the hoisting dynamic load coefficient of  $\phi_{2\min}$  —— corresponding to the hoisting status level is 1.1.

$\beta_2$  ——按起升状态级别设定的系数, 取为 0.34;

$\beta_2$  - Coefficient set according to the level of the hoisting status: 0.34;

$v_q$  ——稳定起升速度, m/s, 与起升机构驱动控制型式及操作方法有关。

$v_q$  ——The stable hoisting speed (m/s) is related to the driving control type and operation method of the hoisting mechanism.

(3) 运行冲击系数  $\phi_4$  :

(3) Operating impact coefficient  $\phi_4$  :

起重机或起重机的部分装置, 在不平的道路或轨道上运行时所发生的垂直冲击动力效应, 见图 2.1。

The vertical impact dynamic effect of crane or partial device of the crane during the travelling on the uneven road or track is shown in Fig. 2.1.

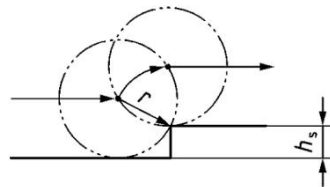


图 2.1 运行中高低差  
Fig. 2.1 Height Difference in Travelling

根据 BS EN 13001-2 中 4.2.2.4 条款计算如下。

The calculation is as follows in accordance with 4.2.2.4 of BS EN 13001-2.

频率  $f_q = 10\text{Hz}$ , 轨道高低差  $h_s = 1\text{mm}$ , 大车车轮半径  $r_y = 400\text{mm}$ , 小车车轮半径  $r_{yt} = 315\text{mm}$ , 则

The frequency is  $f_q = 10\text{Hz}$ , height-low difference of track is  $h_s = 1\text{mm}$ , the crane wheel radius is  $r_y = 400\text{mm}$ , the trolley wheel radius is  $r_{yt} = 315\text{mm}$ ,

$$\text{大车运行时 } \phi_4 = 1 + \left(\frac{\pi}{2}\right)^2 \frac{v_y^2}{gr} \xi_s = 1 + \left(\frac{\pi}{2}\right)^2 \frac{(70/60)^2}{9.81 \times 0.2} \times 0.225 = 1.39$$

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$\phi_4 = 1 + \left(\frac{\pi}{2}\right)^2 \frac{v_y^2}{gr} \xi_s = 1 + \left(\frac{\pi}{2}\right)^2 \frac{(70/60)^2}{9.81 \times 0.2} \times 0.225 = 1.39$  during the operation of the crane

小车运行时  $\phi_4 = 1 + \left(\frac{\pi}{2}\right)^2 \frac{(55/60)^2}{9.81 \times 0.1575} \times 0.354 = 1.475$

$\phi_4 = 1 + \left(\frac{\pi}{2}\right)^2 \frac{(55/60)^2}{9.81 \times 0.1575} \times 0.354 = 1.475$  during the operation of the trolley

(4) 动载系数  $\phi_5$

(4) Dynamic load coefficient  $\phi_5$

由驱动机构加速或减速、起重机意外停机或传动机构突然失效等原因在起重机中引起的弹性效应，取为 1.5。

The elastic effect in the crane caused by the acceleration or deceleration of the driving mechanism, the unplanned shutdown of the crane or the sudden failure of the transmission mechanism is 1.5.

### 2.2.2. 桥架自重载荷 DL

### 2.2.2 Self-weight Load DL of Bridge frame

主要有：桥架自重（包括有主梁、轨道、走台、栏杆、电气及导电装置等）、大车运行机构、操纵司机室以及电气室等重量。

The self-weight of the bridge frame (including the main beam, trolley rail, platform, railings, electrical and conductive devices etc.), and the weight of the crane travelling mechanism, the operator cabin and the electrical room are mainly included.

表 2.3 自重载荷

Table 2.3 Self-weight Load

| 项目<br>Project                       | 重量<br>Weight | 单位<br>Unit |
|-------------------------------------|--------------|------------|
| 桥架<br>bridge frame                  | 39.2         | 吨<br>Ton   |
| 电气<br>Electrical                    | 5.4          | 吨<br>Ton   |
| 大车运行机构<br>Crane traveling mechanism | 3.2          | 吨<br>Ton   |
| 总和<br>Sum                           | 47.8         | 吨<br>Ton   |

桥架及其上部件的自重以均布载荷考虑， $q_G = 44.6 \times 9.81 / 37.1 / 2 = 5.90 \text{ kN/m}$ （单根主梁）。

The self-weight of the crane bridge frame and the components thereof is considered as the uniform distribution load,  $q_G = 44.6 \times 9.81 / 37.1 / 2 = 5.90 \text{ kN/m}$  (single main beam).

签字：  
Signature:  
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大车运行机构的自重作用在桥架的支座处。

The self-weight of the crane travelling mechanism acts on the support of the bridge frame.

### 2.2.3 移动载荷 TL+LS+LL

### 2.2.3 Moving Load TL+LS+LL

主梁上的移动载荷由起升机构、小车的自重和起升载荷组成。起升载荷是由起升质量引起的重力。

The moving load on the main beam is composed of the self-weight of the hoisting mechanism, trolley mechanism and the hoisting load. The hoisting load is the gravity caused by the hoisting mass.

表 2.4 移动载荷

Table 2.4 Moving Load

| 项目<br>Project                        |                                                             | 重量<br>Weight | 单位<br>Unit |
|--------------------------------------|-------------------------------------------------------------|--------------|------------|
| 小车自重 TL<br>Self-weight of trolley TL |                                                             | 10.8         | 吨<br>Ton   |
| 起升载荷<br>Hoisting load                | 电动抓斗 LS<br>Motor driving grab LS                            | 7.8          | 吨<br>Ton   |
|                                      | 起重量（包括抓斗）LS+LL<br>Hoisting weight (including grab)<br>LS+LL | 18.0         | 吨<br>Ton   |
|                                      |                                                             |              |            |

不考虑动载效应的移动载荷  $\Sigma P = (10.8+18) \times 9.81 = 282.528 \text{ kN}$

Moving load without considering the dynamic load effect:  $\Sigma P = (10.8+18) \times 9.81 = 282.528 \text{ kN}$

根据简化假定，小车轮压按四个均等的集中力考虑，则

According to the simplified assumption, the trolley wheel load shall be considered as four uniform concentrated force,

$$P_d = \frac{\phi_2 \cdot (LL + LS) + \phi_4 \cdot TL}{4} = (1.38 \times 18 + 1.39 \times 10.8) \times 9.81 / 4 = 97.74 \text{ kN}$$

### 2.2.4 运行惯性载荷 LATG/LATT

### 2.2.4 Travelling Inertial Load LATG/LATT

根据刚体动力学原理进行计算，还需考虑结构的动载效应，即乘以  $\phi_5$  加以考虑。

按打滑条件，计算公式如下所示：

According to the principle of rigid body dynamics, the dynamic load effect of the structure shall also be considered, that is, it is multiplied by  $\phi_5$ . According to the slipping conditions, the calculation formula is as follows:

$$LATT = ma_{yt} < \mu P_L, \quad LATG = ma_y < \mu P_L$$

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式中：  $\mu$  ——系数，当运行机构总轮数的 1/2 为驱动轮时，取  $\mu=0.1$ （室内）；  
In the formula:  $\mu$  - coefficient, when half of the total number of the wheels in the operation mechanism refers to the driving wheel,  $\mu=0.1$  (indoor);

$P_L$  ——小车车轮上的总垂直静轮压；

$P_L$  - The total vertical static wheel pressure on the trolley wheel;

$q$  ——桥架自重载荷。

$q$  - Self-weight load of bridge frame

$$LATT = 0.1 \times (10.8 + 18) \times 9.81 = 28.26 \text{ kN}$$

$$LATG = 0.1 \times 44.6 \times 9.81 = 43.75 \text{ kN}$$

小车运行机构起动或制动时，由移动载荷引起的惯性载荷是以集中力  $P_H$  的方式沿水平方向作用在跨度中或端部，并通过小车轮缘与轨道侧向接触从而传递给主梁；大车运行机构起动或制动时，桥架部分质量引起的水平惯性载荷是以均布载荷  $q_H$  的方式作用在主梁上。 $P_H$  和  $q_H$  值可以表达为

When the trolley travelling mechanism is in the starting or braking state, the inertial load caused by the moving load acts in the span or at the end in the horizontal direction by means of the concentrated force  $P_H$ , and is transmitted to the main beam through the lateral contact between the trolley rim and the track. When the crane travelling mechanism is in the starting or braking status, the horizontal inertial load caused by part of the mass of the bridge frame is applied to the main beam by means of the uniform load  $q_H$ .  $P_H$  The value  $q_H$  can be expressed as follows:

$$P_H = \phi_3 LATT / 4 = 1.5 \times 28.26 / 4 = 10.60 \text{ kN}$$

$$q_H = \phi_3 \frac{LATG}{L} = 1.5 \times 43.75 / 37.1 / 2 = 0.884 \text{ kN/m}$$

当满载小车制动时，它将产生沿主梁轨道方向的水平惯性载荷  $P'_H$ 。按打滑条件，作用在一根主梁上的  $P'_H$  可按下面公式计算：

During the trolley is being braked with full load, it will produce the horizontal inertial load  $P'_H$  along the direction of the main beam rail. According to the slipping condition,  $P'_H$  effecting on the main beam can be calculated according to the following formula:

$$P_{dy} = \frac{\phi_4 TL + \phi_2 (LS + LL)}{4} = (1.153 \times 10.8 + 1.38 \times 18) \times 9.81 / 4 = 91.46 \text{ kN}$$

$$P'_H = \mu P_{dy} = 0.1 \times 91.46 = 9.146 \text{ kN}$$

式中：  $P_{dy}$  ——作用在一根主梁上的小车驱动轮轮压之和；

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姓名：彭军

Name: Peng Jun

日期：2024 年 12 月 24 日

Date: December 24, 2024

职务：机械工程师

Position: Mechanical Engineer

In the formula:  $P_{dy}$  - the sum of the wheel load of the trolley driving wheel effecting on the main beam;

$\phi_2, \phi_4$ ——起升载荷动载系数和运行冲击系数。

$\phi_2, \phi_4$  - Hoisting load and dynamic load coefficient and operating impact coefficient.

### 2.2.5 试验载荷 $P_t$

### 2.2.5 Test Load $P_t$

静载试验载荷取为 1.25 起升载荷，动态试验载荷取为 1.1 起升载荷。在验算时，动态试验载荷应乘以试验载荷起升动载系数  $\phi_6 = \phi_2 = 1.38$ 。

The static load test load is the hoisting load of 1.25, and the dynamic test load is the hoisting load of 1.1. In the checking calculation, the dynamic test load shall be multiplied by the test load and hoisting dynamic load coefficient of  $\phi_6 = \phi_2 = 1.38$ .

### 2.2.6 偏斜运行时的水平侧向力 $P_s$

### 2.2.6 Horizontal Lateral Force during the Eccentric Travelling $P_s$

大车偏斜侧向力  $P_s$  是以一对力偶的形式作用在同一侧端梁的两个车轮上，或作用在桥架的对角线位置的两个轮上，见图 2.2。

The eccentric lateral force  $P_s$  of the crane is in the form of a pair of couples effecting on the two wheels of the end beam on the same side, or on the two wheels of the diagonal position of the bridge frame. See Fig. 2.2.

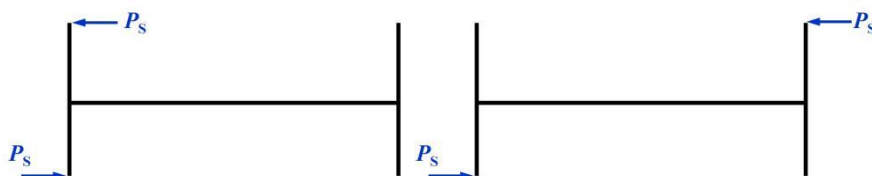


图 2.2 大车偏斜运行时的水平侧向力作用图

Fig. 2.2 Horizontal Lateral Force Action during Crane Eccentric Travelling

根据 BS EN 13001-2 中 4.2.3.4 条款计算如下：

The calculation is as follows in accordance with 4.2.3.4 of BS EN 13001-2:

轨道采用宽度为  $b_h = 70\text{mm}$  的方钢，起重机单侧轨道上车轮数量  $n = 4$ 。

The rail adopts the square steel with the width of  $b_h = 70\text{mm}$ , and the number of wheels on the single side rail of the crane is  $n = 4$ .

轮对  $i$  与导向装置的距离  $d_i = 0.245, 2.245, 3.445, 5.445\text{m}$ ，前后导向轮之间的距离  $W_b = 5.69\text{m}$ 。按偏斜角  $\alpha = 0.01\text{rad}$ ，则

The distance between the wheelset  $i$  and the guide device is  $d_i = 0.245, 2.245, 3.445, \text{ and } 5.445\text{m}$ . The distance between the front and rear guide wheels is  $W_b = 5.69\text{m}$ . If

签字：

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Company Stamp:

姓名：彭军

Name: Peng Jun

日期：2024 年 12 月 24 日

Date: December 24, 2024

职务：机械工程师

Position: Mechanical Engineer



the declination angle is  $\alpha = 0.01\text{rad}$ ,

$$\text{系数 } f = \mu_0 \times [1 - e^{(-250\alpha)}] = 0.3 \times (1 - e^{-2.5}) = 0.275$$

$$\text{the coefficient is } f = \mu_0 \times [1 - e^{(-250\alpha)}] = 0.3 \times (1 - e^{-2.5}) = 0.275.$$

在单个车轮上的平均偏斜载荷  $P_s = v \times f \times mg / n = 0.9524 \times 0.275 \times (47.8 + 10.8 + 18) \times 9.81 / 8 = 23.94\text{kN}$

The average deflection load on single wheel is  $P_s = v \times f \times mg / n = 0.9524 \times 0.275 \times (47.8 + 10.8 + 18) \times 9.81 / 8 = 23.94\text{kN}$ .

### 3.主梁的计算简图和计算载荷的确定

### 3. Calculation diagram of main beam and determination of calculation load

#### 3.1.主梁的截面参数

#### 3.1 Sectional Parameters of Main Beam

主梁的截面如图 3.1 所示，坐标系如图 3.2 所示，截面参数如表 3.1。

The section of the main beam is as shown in Fig. 3.1, the coordinate system is as shown in Fig. 3.2, and the sectional parameters are as shown in Table 3.1.

表 3.1 截面参数表

Table 3.1 Sectional Parameters

| 截面<br>Section                                                                                | 主截面<br>Main Section | 与端梁连接截面<br>Selection<br>Connected with the<br>End Beam |                                    |
|----------------------------------------------------------------------------------------------|---------------------|--------------------------------------------------------|------------------------------------|
| 梁高, mm<br>Beam height, mm                                                                    | 1720                | 800                                                    |                                    |
| 距中性轴最大边缘距离 $y_{\max}$ , mm<br>Maximum edge distance from the neutral<br>axis $y_{\max}$ , mm | 442.7               | 433.0                                                  | 水平面内<br>In the horizontal<br>plane |
| 距中性轴最大边缘距离 $z_{\max}$ , mm<br>Maximum edge distance from the neutral<br>axis $z_{\max}$ , mm | 915.7               | 440.9                                                  | 铅锤面内<br>In the plumb<br>plane      |
| 翼缘面积 $A_y$ , mm <sup>2</sup><br>Flange area $A_y$ , mm <sup>2</sup>                          | 2.0980E+04          | 2.0980E+04                                             |                                    |
| 腹板面积 $A_z$ , mm <sup>2</sup><br>Web area $A_z$ , mm <sup>2</sup>                             | 2.4080E+04          | 1.1200E+04                                             |                                    |
| 扇形惯性极 $2\Omega$ , mm <sup>2</sup><br>Sector inertia pole $2\Omega$ , mm <sup>2</sup>         | 2.49E+06            | 1.17E+06                                               |                                    |

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Company Stamp:

姓名: 彭军

Name: Peng Jun

日期: 2024 年 12 月 24 日

Date: December 24, 2024

职务: 机械工程师

Position: Mechanical Engineer

|                                                                                 |            |            |  |
|---------------------------------------------------------------------------------|------------|------------|--|
| 轴惯性矩 $I_y$ , mm <sup>4</sup><br>Axial moment of inertia $I_y$ , mm <sup>4</sup> | 2.1605E+10 | 4.0385E+09 |  |
| 轴惯性矩 $I_z$ , mm <sup>4</sup><br>Axial moment of inertia $I_z$ , mm <sup>4</sup> | 4.2198E+09 | 2.5510E+09 |  |
| 截面模量 $W_{y1}$ , mm <sup>3</sup><br>Section modulus $W_{y1}$ , mm <sup>3</sup>   | 2.6020E+07 | 1.0486E+07 |  |
| 截面模量 $W_{y2}$ , mm <sup>3</sup><br>Section modulus $W_{y2}$ , mm <sup>3</sup>   | 2.3594E+07 | 9.1602E+06 |  |
| 截面模量 $W_{z1}$ , mm <sup>3</sup><br>Section modulus $W_{z1}$ , mm <sup>3</sup>   | 1.2652E+07 | 7.5407E+06 |  |
| 截面模量 $W_{z2}$ , mm <sup>3</sup><br>Section modulus $W_{z2}$ , mm <sup>3</sup>   | 9.9722E+06 | 6.2571E+06 |  |

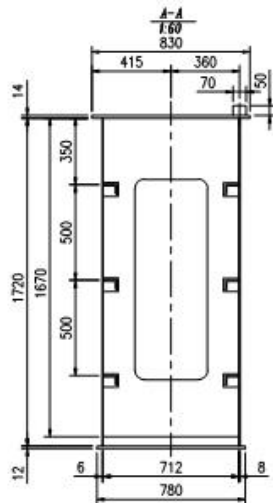


图 3.1 主梁截面  
Fig. 3.1 Main Beam Section

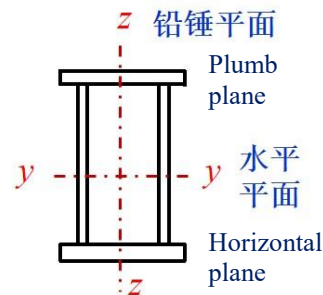


图 3.2 截面坐标系  
Fig. 3.2 Sectional Coordinate System

### 3.2.主梁的计算简图 3.2 Calculation Diagram of Main Beam

主梁强度计算简图如图 3.2 所示，图中给出了桥架在铅锤和水平平面内所承受的载荷，以及小车作用的位置。

The strength calculation diagram of the main beam is as shown in Fig. 3.2, which shows the load borne by the bridge frame in the plumb and horizontal planes, and the action position of the trolley.

签字:  
Signature:  
公司签章:  
Company Stamp:

姓名: 彭军  
Name: Peng Jun  
日期: 2024 年 12 月 24 日  
Date: December 24, 2024

职务: 机械工程师  
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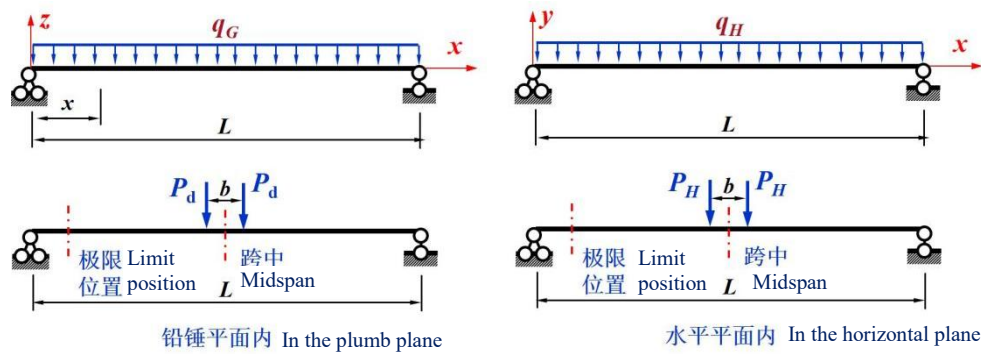


图 3.2 主梁的计算简图

Fig. 3.2 Calculation Diagram of Main Beam

由于本方案采用的是偏轨梁，因此还存在扭矩，偏轨的距离为 360mm。

As the deflection beam is adopted in this scheme, there is still torque, and the deflection distance is 360mm.

### 3.3.主梁的强度验算原理

### 3.3 Strength Checking Principle of Main Beam

强度计算的截面为：当小车位于跨中时的 I-I 和 II-II 截面，以及小车位于极限位置时的 III-III 截面。各截面所考虑的

The sections for strength calculations are I-I and II-II sections when the trolley is in the middle of span, and III-III section when the trolley is at the limit position. Considered for each section

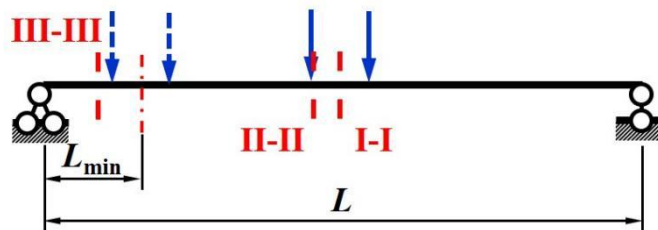


图 3.3 计算截面

Fig. 3.3 Calculation Section

(1) I-I 截面，由于垂直方向和水平方向最大弯矩同时作用，在主梁截面的翼缘板中引起的最大正应力为：

(1) For the I-I section, due to the simultaneous action of the maximum bending moments in the vertical and horizontal directions, the maximum normal stress caused in the flange plate of the main beam section is:

$$\sigma = \frac{\sum M_y}{W_y} + \frac{\sum M_z}{W_z} \leq [\sigma] \quad (10-11)$$

式中： $\sum M_y, \sum M_z$ ——分别为由垂直载荷和水平载荷在跨中引起的最大弯矩；

签字：  
Signature:  
公司签章：  
Company Stamp:

姓名：彭军  
Name: Peng Jun  
日期：2024 年 12 月 24 日  
Date: December 24, 2024

职务：机械工程师  
Position: Mechanical Engineer



In the formula:  $\sum M_y, \sum M_z$  - the maximum bending moment in span caused by vertical load and horizontal load respectively;

(2) II-II 截面, 主梁的截面内同时存在弯矩和剪力的作用, 应验算截面中由弯曲正应力  $\sigma$  和剪应力  $\tau$  产生的复合应力, 可按下式计算:

(2) There are both bending moment and shearing force in the section of II-II and the section of the main beam. The combined stress generated by the normal stress  $\sigma$  and shearing stress  $\tau$  in bending in the section shall be checked, which can be calculated according to the following formula:

$$\sqrt{\sigma^2 + 3\tau^2} \leq [\sigma] \quad (10-12)$$

式中:  $\sigma$  ——由垂直载荷和水平载荷在 II-II 截面危险点处引起的弯矩正应力;  
In the formula:  $\sigma$  - the normal stress in bending caused by vertical and horizontal loads at the danger point of II-II section;

$\tau$  ——由垂直载荷在 II-II 截面的危险点处引起的剪应力。

$\tau$  - Shearing stress caused by vertical load at the danger point of II-II section.

(3) III-III 截面, 主要验算跨端截面由垂直剪力  $Q_y$  引起的腹板最大剪应力, 应满足:

(3) In the III-III section, the checking calculation is mainly performed for the maximum shearing stress of the web caused by vertical shearing force  $Q_y$  across the end section shall meet the following requirements:

$$\tau = \frac{Q_z S_y(\omega)}{I_y \sum t_w} \leq [\tau] \quad (10-13)$$

式中:  $Q_z$  ——由垂直载荷引起的计算截面上的最大剪力;

In the formula:  $Q_z$  - maximum shearing force on the calculation section caused by the vertical load;

$S_y(\omega)$  ——计算剪应力处以上截面对水平轴  $y-y$  的静矩;

$S_y(\omega)$  - Calculate the static moment of the section above the shearing stress with respect to the horizontal axis  $y-y$ ;

$\sum t_w$  ——主梁腹板的总厚度, 即为两块腹板的厚度  $2t_w$ ;

$\sum t_w$  - Total thickness of the main beam web, just the thickness of two webs  $2t_w$ ;

$[\tau]$  ——结构设计的控制应力,  $[\tau] = [\sigma] / \sqrt{3}$ 。

$[\tau]$  - Control stress of structural design,  $[\tau] = [\sigma] / \sqrt{3}$ 。

签字:  
Signature:  
公司签章:  
Company Stamp:

姓名: 彭军  
Name: Peng Jun  
日期: 2024 年 12 月 24 日  
Date: December 24, 2024

职务: 机械工程师  
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### 3.4.强度计算结果和校核

### 3.4 Strength Calculation Results and Check

各工况下，危险截面上的应力情况如列于表 3.2。  
The stress conditions on the dangerous section under various working conditions are as shown in Table 3.2.

表 3.2 应力汇总表，MPa  
Table 3.2 Stress Summary, MPa

| 载荷组合<br>Load<br>Combination | 危险截面<br>Dangerous<br>Section  | 最大正应力<br>Maximum<br>Normal Stress | 最大切应力<br>Maximum<br>Shearing Stress | 复合应力<br>Combined<br>Stress | 强度判断<br>Strength<br>Judgment              |
|-----------------------------|-------------------------------|-----------------------------------|-------------------------------------|----------------------------|-------------------------------------------|
| A1                          | 截面 I-I<br>Section I-I         | 113.99                            | 4.34                                | 114.24                     | [ $\sigma$ ]=218<br>[ $\tau$ ]=126<br>OK! |
|                             | 截面 II-II<br>Section II-II     | 113.99                            | 8.40                                | 114.92                     |                                           |
|                             | 截面 III-III<br>Section III-III | 59.52                             | 35.10                               | 85.08                      |                                           |
| A2                          | 截面 I-I<br>Section I-I         | 132.54                            | 4.34                                | 132.75                     |                                           |
|                             | 截面 II-II<br>Section II-II     | 132.54                            | 8.40                                | 133.34                     |                                           |
|                             | 截面 III-III<br>Section III-III | 89.17                             | 35.11                               | 107.93                     |                                           |
| A3                          | 截面 I-I<br>Section I-I         | 150.66                            | 4.71                                | 150.88                     |                                           |
|                             | 截面 II-II<br>Section II-II     | 150.66                            | 9.17                                | 151.49                     |                                           |
|                             | 截面 III-III<br>Section III-III | 98.53                             | 39.99                               | 120.44                     |                                           |
| B1                          | 截面 I-I<br>Section I-I         | 147.69                            | 4.34                                | 147.88                     | [ $\sigma$ ]=240<br>[ $\tau$ ]=139<br>OK! |
|                             | 截面 II-II<br>Section II-II     | 147.69                            | 8.42                                | 148.40                     |                                           |
|                             | 截面 III-III<br>Section III-III | 113.22                            | 35.14                               | 128.54                     |                                           |
| B2                          | 截面 I-I<br>Section I-I         | 152.18                            | 4.71                                | 152.40                     |                                           |
|                             | 截面 II-II<br>Section II-II     | 152.18                            | 9.18                                | 153.01                     |                                           |
|                             | 截面 III-III<br>Section III-III | 100.83                            | 39.99                               | 122.33                     |                                           |
| C1                          | 截面 I-I<br>Section I-I         | 122.63                            | 4.06                                | 122.83                     | [ $\sigma$ ]=264<br>[ $\tau$ ]=152<br>OK! |
|                             | 截面 II-II<br>Section II-II     | 122.63                            | 7.89                                | 123.38                     |                                           |
|                             | 截面 III-III<br>Section III-III | 77.88                             | 33.54                               | 97.16                      |                                           |
| C2                          | 截面 I-I                        | 156.66                            | 5.00                                | 156.90                     |                                           |

签字：  
Signature:  
公司签章：  
Company Stamp:

姓名：彭军  
Name: Peng Jun  
日期：2024 年 12 月 24 日  
Date: December 24, 2024

职务：机械工程师  
Position: Mechanical Engineer



|  |                 |        |       |        |  |
|--|-----------------|--------|-------|--------|--|
|  | Section I-I     |        |       |        |  |
|  | 截面 II-II        | 156.66 | 9.72  | 157.57 |  |
|  | Section II-II   |        |       |        |  |
|  | 截面 III-III      | 103.16 | 41.65 | 125.88 |  |
|  | Section III-III |        |       |        |  |

### 3.5.刚度校核

### 3.5 Rigidity Check

仅考虑移动载荷引起的下挠度。

The down deflection caused by the moving load shall be considered only.

$$\alpha = (L-b)[3L^2 - (L-b)^2] / 2L^3 = \frac{(37.1-2.4)[3 \times 37.1^2 - (37.1-2.4)^2]}{2 \times 37.1^3} = 0.994$$

$$\text{静刚度 } Y_L = \frac{\alpha \Sigma PL^3}{48EI} = \frac{0.991 \times 282.528 \times 10^3 / 2 \times 37.1^3 \times 10^9}{48 \times 2.01 \times 10^5 \times 2.1605 \times 10^{10}} = 34.3 \text{ mm}$$

$$\text{Static rigidity } Y_L = \frac{\alpha \Sigma PL^3}{48EI} = \frac{0.991 \times 282.528 \times 10^3 / 2 \times 37.1^3 \times 10^9}{48 \times 2.01 \times 10^5 \times 2.1605 \times 10^{10}} = 34.3 \text{ mm}$$

许用静刚度 $[Y_L]=L/1000=37.1\text{mm}$  静刚度满足要求！

Allowable static rigidity:  $[Y_L]=L/1000=37.1\text{mm}$  The static rigidity meets the requirements!

## 4.端梁的设计和计算

## 4. Design and Calculation of End Beam

### 4.1.计算简图和作用载荷

### 4.1 Calculation Diagram and Action Load

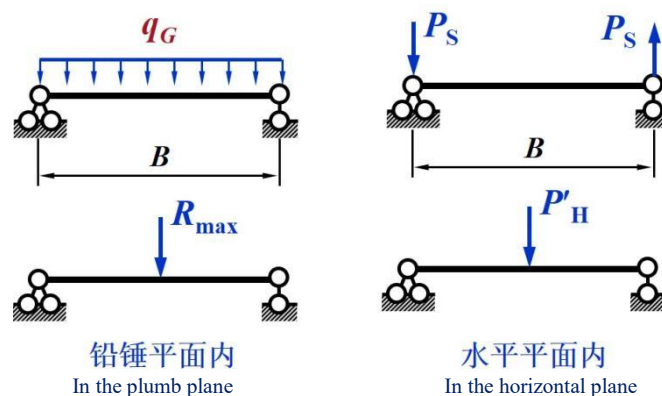


图 4.1 端梁的计算简图

Fig. 4.1 Calculation Diagram of End Beam

签字:  
Signature:  
公司签章:  
Company Stamp:

姓名: 彭军  
Name: Peng Jun  
日期: 2024 年 12 月 24 日  
Date: December 24, 2024

职务: 机械工程师  
Position: Mechanical Engineer

端梁的计算简图如图 4.1 所示。作用在端梁上的载荷有自重载荷  $q_G$ 、主梁的支座反力  $R_{\max}$ （小车在极限位置时主梁的最大轮压，见表 4.1）、小车运行的惯性载荷  $P'_H$  和作用在大车运行机构上的偏斜载荷  $P_S$ 。以上均需考虑动载效应。

The calculation diagram of end beam is as shown in Fig. 4.1. The load effecting on the end beam includes the self-weight load  $q_G$ , reaction force at support of main beam  $R_{\max}$  (maximum wheel load of the main beam when the trolley is at the limit position, see Table 4.1), inertial load  $P'_H$  during trolley travelling and eccentric load  $P_S$  effecting on the crane travelling mechanism. The dynamic load effect shall be considered for all the above.

表 4.1 各工况下的支座反力  $R_{\max}$ , kN

Table 4.1 Reaction at Support under Each Working Condition  $R_{\max}$ , kN

| 载荷工况<br>Load Conditions           | A1      | A2      | A3      | B1      | B2      | C1      | C2      |
|-----------------------------------|---------|---------|---------|---------|---------|---------|---------|
| 支座反力<br>Reaction force at Support | 291.450 | 291.450 | 337.767 | 291.450 | 337.767 | 280.545 | 349.343 |

## 4.2.端梁的截面参数

## 4.2 Sectional Parameters of End Beam

端梁的截面形状和截面参数分别见图 4.2 和表 4.2。

The sectional shape and parameters of the end beam are shown in Fig. 4.2 and Table 4.2 respectively.

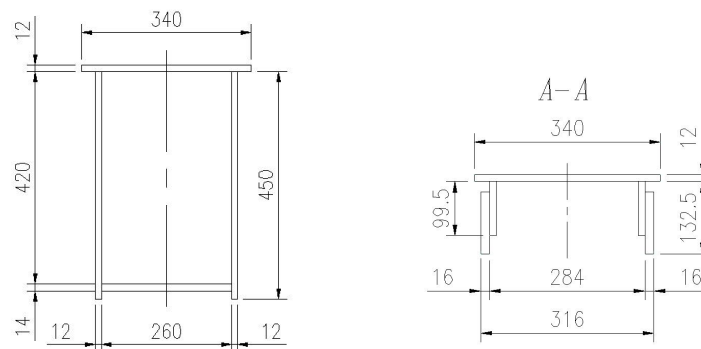


图 4.2 端梁截面

Fig. 4.2 End Beam Section

表 4.2 截面参数表

Table 4.2 Sectional Parameters

| 截面<br>Section | 主截面<br>Main Section | 与端梁连接截面<br>Selection<br>Connected with the<br>End Beam |  |
|---------------|---------------------|--------------------------------------------------------|--|
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签字:

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姓名: 彭军

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|                                                                                           |          |          |                                 |
|-------------------------------------------------------------------------------------------|----------|----------|---------------------------------|
| 梁高, mm<br>Beam height, mm                                                                 | 420      | 132.5    |                                 |
| 距中性轴最大边缘距离 $y_{\max}$ , mm<br>Maximum edge distance from the neutral axis $y_{\max}$ , mm | 260      | 260      | 水平面内<br>In the horizontal plane |
| 距中性轴最大边缘距离 $z_{\max}$ , mm<br>Maximum edge distance from the neutral axis $z_{\max}$ , mm | 12       | 16       | 铅锤面内<br>In the plumb plane      |
| 翼缘面积 $A_y$ , mm <sup>2</sup><br>Flange area $A_y$ , mm <sup>2</sup>                       | 238.94   | 69.23    |                                 |
| 腹板面积 $A_z$ , mm <sup>2</sup><br>Web area $A_z$ , mm <sup>2</sup>                          | 170      | 170      |                                 |
| 扇形惯性极 $2\Omega$ , mm <sup>2</sup><br>Sector inertia pole $2\Omega$ , mm <sup>2</sup>      | 7.72E+03 | 5.44E+03 |                                 |
| 轴惯性矩 $I_y$ , mm <sup>4</sup><br>Axial moment of inertia $I_y$ , mm <sup>4</sup>           | 1.08E+04 | 3.18E+03 |                                 |
| 轴惯性矩 $I_z$ , mm <sup>4</sup><br>Axial moment of inertia $I_z$ , mm <sup>4</sup>           | 2.36E+05 | 1.30E+04 |                                 |
| 截面模量 $W_{y1}$ , mm <sup>3</sup><br>Section modulus $W_{y1}$ , mm <sup>3</sup>             | 5.25E+08 | 1.14E+07 |                                 |
| 截面模量 $W_{y2}$ , mm <sup>3</sup><br>Section modulus $W_{y2}$ , mm <sup>3</sup>             | 2.60E+08 | 8.35E+07 |                                 |
| 截面模量 $W_{z1}$ , mm <sup>3</sup><br>Section modulus $W_{z1}$ , mm <sup>3</sup>             | 2.20E+06 | 1.65E+05 |                                 |
| 截面模量 $W_{z2}$ , mm <sup>3</sup><br>Section modulus $W_{z2}$ , mm <sup>3</sup>             | 2.72E+06 | 2.69E+05 |                                 |

### 4.3.强度校核

### 4.3 Strength Check

端梁强度计算部位为: ⊕ 弯矩最大且安装拼接截面  $I-I$  (主梁与端梁的连接处);  
⊗ 支承截面  $II-II$ , 见图 4.3。

The calculation position of the end beam strength is: ⊕ Maximum bending moment and installed splicing section:  $I-I$  (connection between the main beam and the end beam); ⊗ Supporting section:  $II-II$ . See Fig. 4.3.

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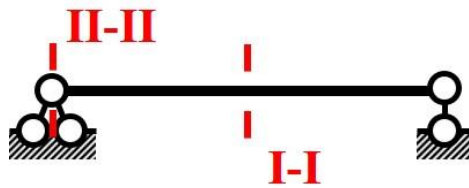


图 4.3 端梁维修截面示意图

Fig. 4.3 Schematic Diagram of End Beam Maintenance Section

各工况下，危险截面上的应力情况如表 4.3。

The stress conditions on the dangerous section under various working conditions are as shown in Table 4.3.

表 4.3 应力汇总表，MPa

Table 4.3 Stress Summary, MPa

| 载荷组合<br>Load<br>Combination | 危险截面<br>Dangerous<br>Section | 最大正应力<br>Maximum<br>Normal Stress | 最大切应力<br>Maximum<br>Shearing Stress | 合成应力<br>Resulting Stress | 强度判断<br>Strength<br>Judgment              |
|-----------------------------|------------------------------|-----------------------------------|-------------------------------------|--------------------------|-------------------------------------------|
| A1                          | 截面 I-I<br>Section I-I        | 42.63                             | 14.88                               | 49.81                    | [ $\sigma$ ]=218<br>[ $\tau$ ]=126<br>OK! |
|                             | 截面 II-II<br>Section II-II    | 0.00                              | 85.49                               |                          |                                           |
| A2                          | 截面 I-I<br>Section I-I        | 66.82                             | 23.43                               | 78.18                    |                                           |
|                             | 截面 II-II<br>Section II-II    | 0.00                              | 50.14                               |                          |                                           |
| A3                          | 截面 I-I<br>Section I-I        | 77.49                             | 27.18                               | 90.67                    |                                           |
|                             | 截面 II-II<br>Section II-II    | 0.00                              | 58.18                               |                          |                                           |
| B1                          | 截面 I-I<br>Section I-I        | 75.64                             | 26.24                               | 88.25                    | [ $\sigma$ ]=240<br>[ $\tau$ ]=139<br>OK! |
|                             | 截面 II-II<br>Section II-II    | 0.00                              | 108.48                              |                          |                                           |
| B2                          | 截面 I-I<br>Section I-I        | 93.93                             | 27.18                               | 105.07                   |                                           |
|                             | 截面 II-II<br>Section II-II    | 0.00                              | 58.18                               |                          |                                           |
| C1                          | 截面 I-I<br>Section I-I        | 70.32                             | 24.42                               | 82.06                    | [ $\sigma$ ]=264<br>[ $\tau$ ]=152<br>OK! |
|                             | 截面 II-II<br>Section II-II    | 0.00                              | 87.58                               |                          |                                           |
| C2                          | 截面 I-I<br>Section I-I        | 105.54                            | 30.92                               | 118.35                   |                                           |
|                             | 截面 II-II<br>Section II-II    | 0.00                              | 119.29                              |                          |                                           |

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## 5.主梁和端梁螺栓连接的校核

### 5. Check of Bolt Connection between Main Beam and End Beam

#### 5.1.计算载荷

#### 5.1 Calculation Load

螺栓群受到的内力是剪力（小车、大车运行惯性引起 $Q_x$ 、 $Q_y$ 、 $M_z$ ）、压力（小车移动到极限位置时的支反力 $N_z$ ）、弯矩（小车运行惯性引起 $M_y$ ，其作用面高于螺栓连接面；偏轨布置的引起 $M_x$ ），其中弯矩会引起外排螺栓受拉。单根端梁上的螺栓布置见图 5.1。

The internal force of the bolt group refers to the shearing force ( $Q_x$ ,  $Q_y$  and  $M_z$  caused by the travelling inertia of the trolley and crane), the pressure (reaction force  $N_z$  when the trolley is moved to the limit position), and the bending moment ( $M_y$  caused by the operating inertia of the trolley, with the effecting surface higher than the connecting surface of the bolt;  $M_x$  caused by deviation arrangement), wherein the outer row of bolts will be strained due to the bending moment. The bolt arrangement on single end beam is as shown in Fig. 5.1.

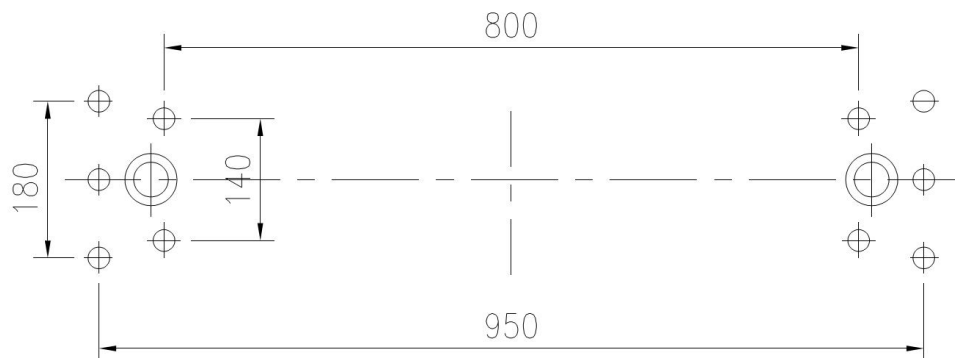


图 5.1 螺栓群的布置图

Fig. 5.1 Layout of Bolt Group

外排螺栓布置间距不能超过 $8d$ 或 $12\delta_{\min}$ 中的最小值，即 $12 \times 12 = 144\text{mm}$ ；最小间距不能小于 $3d = 3 \times 24 = 72\text{mm}$ ，满足布置要求。

The distance between outer row of bolts cannot exceed the minimum value in  $8d$  or  $12\delta_{\min}$ ,  $12 \times 12 = 144\text{mm}$ ; the minimum distance shall not be less than  $3d = 3 \times 24 = 72\text{mm}$ , meeting the layout requirements.

螺栓群的惯性矩

Inertia moment of bolt group

$$I_y = \sum n_i x_i^2 = 4 \times (70^2 + 90^2) = 0.52 \times 10^5 \text{ mm}^2$$

签字:

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Company Stamp:

姓名: 彭军

Name: Peng Jun

日期: 2024 年 12 月 24 日

Date: December 24, 2024

职务: 机械工程师

Position: Mechanical Engineer



$$I_x = \sum n_i y_i^2 = 4 \times 400^2 + 6 \times 475^2 = 1.99375 \times 10^6 \text{ mm}^2$$

小车运行轨道顶面到螺栓接触面的高度  $h = 884 \text{ mm}$

Height between the top surface of the trolley track and the contact surface of the bolt:  
 $h = 884 \text{ mm}$

小车轨道的偏心距离  $b = 330 \text{ mm}$

Eccentric distance of the trolley track:  $b = 330 \text{ mm}$

$$\text{剪力 } Q_x = \phi_5 \text{ LATT} = 1.5 \times (28.8 \times 0.3) / 2 = 6.48 \text{ kN}$$

$$\text{Shearing force } Q_x = \phi_5 \text{ LATT} = 1.5 \times (28.8 \times 0.3) / 2 = 6.48 \text{ kN}$$

$$\text{剪力 } Q_y = \phi_5 \text{ LATG} = 1.5 \times (73.4 \times 0.3) / 2 = 16.515 \text{ kN}$$

$$\text{Shearing force } Q_y = \phi_5 \text{ LATG} = 1.5 \times (73.4 \times 0.3) / 2 = 16.515 \text{ kN}$$

$$\text{压力 } N_z = R_{\max} / 2 = 349.34 / 2 = 174.67 \text{ kN}$$

$$\text{Pressure } N_z = R_{\max} / 2 = 349.34 / 2 = 174.67 \text{ kN}$$

$$\text{弯矩 } M_y = Q_x h = 6.48 \times 0.884 = 5.73 \text{ kN} \cdot \text{m} \text{ (小车惯性引起)}$$

$$\text{Bending moment } M_y = Q_x h = 6.48 \times 0.884 = 5.73 \text{ kN} \cdot \text{m} \text{ (caused by trolley inertia)}$$

$$\text{扭矩 } T_z = Q'_x L_{\min} = 1.5 \times 28.8 \times 0.3 / 2 \times 1.85 =$$

$$\text{Torque } T_z = Q'_x L_{\min} = 1.5 \times 28.8 \times 0.3 / 2 \times 1.85 =$$

$$\text{弯矩 } M_x = R_{\max} b = 174.67 \times 0.330 = 57.64 \text{ kN} \cdot \text{m} \text{ (轮压由于偏轨引起)}$$

$$\text{Bending moment: } M_x = R_{\max} b = 174.67 \times 0.330 = 57.64 \text{ kN} \cdot \text{m} \text{ (the wheel load is caused by rail offset)}$$

对于螺栓 1，弯矩引起的拉力为

For bolt 1, the tensile force caused by the bending moment is

$$N_{zt}^1 = \frac{M_y}{I_y} x_1 + \frac{M_x}{I_x} y_1$$

$$= 5.73 \times 10^6 / (0.392 \times 10^5 / 0.4) + 57.64 \times 10^6 / (1.4681 \times 10^6 / 0.09) = 62.00 \text{ kN}$$

对于螺栓 2，弯矩引起的拉力为

For bolt 2, the tensile force caused by the bending moment is

$$N_{zt}^2 = \frac{M_y}{I_y} x_2 + \frac{M_x}{I_x} y_2$$

$$= 5.73 \times 10^6 / (0.392 \times 10^5 / 0.455) + 57.64 \times 10^6 / (1.4681 \times 10^6 / 0.065) = 55.435 \text{ kN}$$

单个螺栓的轴向合力为

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姓名: 彭军

Name: Peng Jun

日期: 2024 年 12 月 24 日

Date: December 24, 2024

职务: 机械工程师

Position: Mechanical Engineer



Resultant axial force of single bolt:

$$P_{t1} = N_{zt}^1 - N_z / 8 = 62.0 - 174.67 / 8 = 40.17 \text{ kN} < 0.7 P_g = 0.7 \times 158 = 110.6 \text{ kN}$$

$$P_{t2} = N_{zt}^2 - N_z / 8 = 55.435 - 174.67 / 8 = 33.60 \text{ kN} < 0.7 P_g$$

单个螺栓所承受的剪力为  $P_v = \frac{1}{n} \sqrt{Q_x^2 + Q_y^2} = 2.302 \text{ kN}$

Shearing force on single bolt:  $P_v = \frac{1}{n} \sqrt{Q_x^2 + Q_y^2} = 2.302 \text{ kN}$

## 5.2. 螺栓的许用承载能力

### 5.2 Allowable Bearing Capacity of Bolt

采用 M24 螺栓，螺栓的有效面积  $A_t = 352.5 \text{ mm}^2$ 。

M24 bolt is applied, and the effective area of the bolt is  $A_t = 352.5 \text{ mm}^2$ .

根据 DIN 15018 表 12,  $[\tau'] = 168 \text{ MPa}$ ,  $[\sigma'] = 140 \text{ MPa}$  (5.6 级), 则

According to Table 12 in DIN 15018,  $[\tau'] = 168 \text{ MPa}$ ,  $[\sigma'] = 140 \text{ MPa}$  (Level 5.6),

单个螺栓的许用抗剪能力  $[P_v] = A_t [\tau'] = 352.5 \times 168 \times 10^{-3} = 59.22 \text{ kN}$

Allowable shearing resistance of single bolt:  $[P_v] = A_t [\tau'] = 352.5 \times 168 \times 10^{-3} = 59.22 \text{ kN}$

单个螺栓的许用抗拉能力  $[P_t] = A_t [\sigma'] = 352.5 \times 140 \times 10^{-3} = 49.35 \text{ kN}$

Allowable tensile capacity of single bolt:  $[P_t] = A_t [\sigma'] = 352.5 \times 140 \times 10^{-3} = 49.35 \text{ kN}$

被连接板的材料为 Q355B, 因此螺栓孔壁的承压许用应力  $[\sigma'_c] = 1.4[\sigma] = 1.4 \times 218 = 305.2 \text{ MPa}$ , 则孔壁许用挤压能力

The connecting plate is made of Q355B, so the allowable pressure stress of the bolt hole wall is:  $[\sigma'_c] = 1.4[\sigma] = 1.4 \times 218 = 305.2 \text{ MPa}$ , the allowable extrusion capacity of the hole wall is:

$$[P_c] = d \Sigma \delta [\sigma'_c] = 24 \times 32 \times 305.2 \times 10^{-3} = 234.39 \text{ kN}$$

## 5.3. 螺栓连接校核

### 5.3 Check of Bolt Connection

校核公式为 
$$\sqrt{\left(\frac{P_t}{[P_t]}\right)^2 + \left(\frac{P_v}{[P_v]}\right)^2} \leq 1$$

The check formula is as follows: 
$$\sqrt{\left(\frac{P_t}{[P_t]}\right)^2 + \left(\frac{P_v}{[P_v]}\right)^2} \leq 1$$

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Company Stamp:

姓名: 彭军

Name: Peng Jun

日期: 2024 年 12 月 24 日

Date: December 24, 2024

职务: 机械工程师

Position: Mechanical Engineer



螺栓 1:  $\sqrt{\left(\frac{40.17}{49.35}\right)^2 + \left(\frac{4.883}{59.22}\right)^2} = 0.82 < 1$  满足要求!

Bolt 1:  $\sqrt{\left(\frac{40.17}{49.35}\right)^2 + \left(\frac{4.883}{59.22}\right)^2} = 0.82 < 1$  Meet the requirements!

螺栓 2:  $\sqrt{\left(\frac{28.19}{49.35}\right)^2 + \left(\frac{4.883}{59.22}\right)^2} = 0.333 < 1$  满足要求!

Bolt 2:  $\sqrt{\left(\frac{28.19}{49.35}\right)^2 + \left(\frac{4.883}{59.22}\right)^2} = 0.333 < 1$  Meet the requirements!

$P_v \leq \frac{[P_c]}{1.2} = 234.39/1.2 = 195.3 \text{ MPa}$  满足要求!

$P_v \leq \frac{[P_c]}{1.2} = 234.39/1.2 = 195.3 \text{ MPa}$  Meet the requirements!

## 5.4. 抗剪块的焊缝校核

### 5.4 Check of Weld on Shearing Block

单根端梁设有两个抗剪块为 $\phi 60/\phi 40$ 的定位销(根据图纸),其内圈焊缝为焊高 $h_f = 6 \text{ mm}$ 的单面角焊缝,则焊缝面积 $A_h = 0.7 \times 6 \times \pi \times 40 = 527 \text{ mm}^2$ 。

Single end beam is provided with two positioning pins with the shearing blocks of  $\phi 60/\phi 40$  (according to the drawing), and the inner ring weld thereof is the Single-sided angle weld with the welding height of  $h_f = 6 \text{ mm}$ , then the weld area is  $A_h = 0.7 \times 6 \times \pi \times 40 = 527 \text{ mm}^2$ .

每个抗剪块所承受的剪力为 $4Q$ ,则焊缝的剪应力为

The shearing force borne by each shearing block is  $4Q$ , and the shearing stress of the weld is:

$\tau_h = \frac{4Q}{A_h} = 4 \times 2.302 \times 1000 / 527 = 17.47 \text{ MPa} < [\tau_h]$  满足要求!

$\tau_h = \frac{4Q}{A_h} = 4 \times 2.302 \times 1000 / 527 = 17.47 \text{ MPa} < [\tau_h]$  Meet the requirements!

焊缝剪切许用应力 $[\tau_h] = \frac{[\sigma]}{\sqrt{2}} = \frac{218}{\sqrt{2}} = 154 \text{ MPa}$

Allowable shearing stress of weld:  $[\tau_h] = \frac{[\sigma]}{\sqrt{2}} = \frac{218}{\sqrt{2}} = 154 \text{ MPa}$

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